Sec: SR STAR MODEL-A & APEX, ALL IITS | DT: 05 / 01 /2026
Time: 3 Hrs | GTM – 09 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	A	C	B	C	C	B	C	A	B
11	12	13	14	15	16	17	18	19	20
B	D	C	D	A	B	B	B	D	B
21	22	23	24	25					
39	10	1200	6	6					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	B	C	B	C	B	C	A	C	C
36	37	38	39	40	41	42	43	44	45
B	B	D	A	B	C	D	B	D	C
46	47	48	49	50					
9	42	7	75	3860					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	B	D	A	A	A	C	D	B	A
61	62	63	64	65	66	67	68	69	70
B	B	C	B	A	A	A	C	C	A
71	72	73	74	75					
21	13	100	10	4					

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HINTS
PHYSICS

1. $\frac{ABD}{C}=\frac{(L)(T)(T)}{L}$

$=T^2$

$M^aL^bT^c$

$b=0$

$a=0$

$c=2$

$a+b+c=2$

2. $F_{net}=V\rho g-V\sigma g$

$(V\rho)a_{eff}=Vg(\rho-\sigma)$

$a_{eff}=\left(1-\frac{\sigma}{\rho}\right)g$

$\therefore T=2\pi\sqrt{\frac{l}{\left(1-\frac{\sigma}{\rho}\right)g}}$

$f=\frac{1}{2\pi}\sqrt{\frac{\left(1-\frac{\sigma}{\rho}\right)g}{l}}$

3. Conceptual

4. for proton $\lambda_0=\frac{h}{\sqrt{2mqV}}$

For $\alpha-$particle

$\lambda_\alpha=\frac{h}{\sqrt{2\times 4m\times 2qV}}$

$\Rightarrow \lambda_\alpha=\frac{\lambda_0}{2\sqrt{2}}$

5. From Brewster's law

$\mu=\tan i_p \Rightarrow \frac{c}{v}=\tan 60^0=\sqrt{3}$

$\Rightarrow v=\frac{c}{\sqrt{3}}=\frac{3\times 10^8}{\sqrt{3}}=\sqrt{3}\times 10^8\ m/s$

6. $P=\frac{nE}{t}\Rightarrow 200\times 10^6=\frac{0.1}{100}\ \frac{mc^2}{24\times 60\times 60}$

$\Rightarrow m=192\times 10^{-3}$

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7. Power factor, $\cos\phi = \frac{1}{\sqrt{2}} = \frac{R}{\sqrt{R^2+\omega^2L^2}}$

This gives, $\omega L = R$ when ω is doubled, we find

$$\cos\phi' = \frac{R}{\sqrt{R^2+4\omega^2L^2}} = \frac{R}{\sqrt{R^2+4R^2}} = \frac{1}{\sqrt{5}}$$

8. Inductor can no longer increases magnetic flux density. At saturation $t = \infty$ inductor behave like a conductor

$$L = \frac{\mu_0\mu_0 N^2 A}{l}$$

$$\Rightarrow L\alpha\mu_r$$

$$L\alpha N^2$$

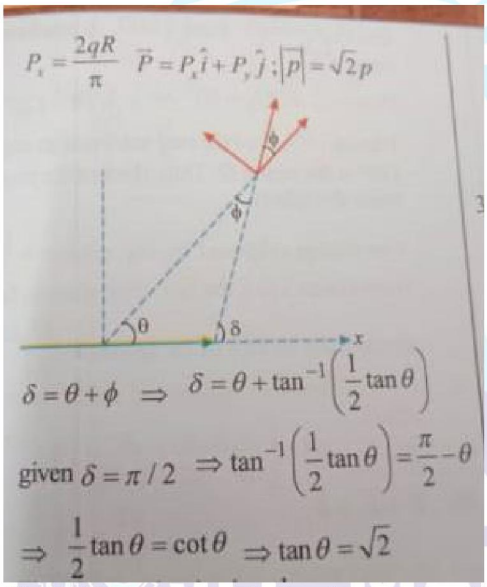
$$L\alpha A$$

$$WD = -\frac{1}{2}LI^2$$

9. Conceptual

10. These options would all be correct, as they reflect the relationships derived from the galvanometer's torque equation $\tau\theta = NIAB$

11.



12. Heat lost by metal ball = heat gained by container and water

$$m_b s_b(500-\theta) = m_c s_c(\theta-30) + m_w s_w(\theta-30)$$

$$0.1\times400(500-\theta) = 800(\theta-30) + 0.5\times4200(\theta-30)$$

$$40(500-\theta) = 2900(\theta-30)$$

$$20000 - 40\theta = 2900\theta - 87000$$

$$\theta = \frac{107000}{2940} = 36.4^0 C$$

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% increase in the temperature of water

$$=\frac{\Delta\theta}{\theta}\times 100=\frac{36.4-30}{30}\times 100=21.33\%$$

13. $$\frac{\sin i}{\sin r}=\mu$$

$$t=\frac{h\sin(i-r)}{\cos r}$$

$$=\frac{10\times\left(\frac{1}{2}\right)}{\cos 30}$$

$$=\frac{10}{\sqrt{3}}cm$$

14. $$V=\sqrt{\frac{2gl\sin\theta}{1+\beta}}$$

15. potential divides across capacitors as

$$\frac{V_1}{V_2}=\frac{C_2}{C_1}=\frac{8}{4}=\frac{2}{1}$$

$$V_1=\frac{2}{3}E=\frac{2}{3}\times 24=16V$$

$$V_2=\frac{1}{3}E=\frac{1}{3}\times 24=8V$$

$$Q_1=C_1V_1$$

$$=4\times 16=64\mu C$$

$$Q_2=C_2V_2$$

$$=8\times 8=64\mu C$$

16. $$F=ax^2+bt^{1/2}$$

$$[a]=\frac{[F]}{[x^2]}=\left[M^1L^{-1}T^{-2}\right]$$

$$[b]=\frac{[F]}{[t^{1/2}]}=\left[M^1L^1T^{-5/2}\right]$$

$$\left[\frac{b^2}{a}\right]=\left[\frac{M^2L^2T^{-5}}{M^1L^{-1}T^{-2}}\right]=\left[M^1L^3T^{-3}\right]$$

17. $$\frac{1}{f_1}=(\mu_1-1)\left(\frac{1}{\infty}+\frac{1}{R}\right)=\frac{\mu_1-1}{R}$$

For the middle lens the focal length is given as

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$$\frac{1}{f_3}=(\mu_3-1)\left(\frac{1}{R}-\frac{1}{R}\right)=0$$

For the plano-concave lens, the focal length is given

$$\frac{1}{f_2}=(\mu_2-1)\left(-\frac{1}{R}+\frac{1}{\infty}\right)=-\frac{(\mu_2-1)}{R}$$

Equivalent focal length of the combination is given as

$$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}$$

$$\Rightarrow\frac{1}{f}=\frac{(\mu_1-1)}{R}-\frac{(\mu_2-1)}{R}$$

$$\Rightarrow\frac{1}{f}=\frac{1}{R}(\mu_1-\mu_2)$$

$$\Rightarrow f=\frac{R}{(\mu_1-\mu_2)}$$

18. $\vec{E}\times\vec{B}$ gives direction of wave propagation

$$\Rightarrow\hat{k}\times\vec{B}\parallel\frac{\hat{i}+\hat{j}}{\sqrt{2}}$$

Now, $\Rightarrow\hat{k}\times\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)=\frac{\hat{j}-(-\hat{i})}{\sqrt{2}}=\frac{\hat{i}-\hat{j}}{\sqrt{2}}$

Wave propagation vector should be along $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

And direction of magnetic field is along $\frac{\hat{i}-\hat{j}}{\sqrt{2}}$

19. Distance from origin means magnitude of the displacement

20. $$F_{net}=4\times\frac{1}{4\pi\in_0}\frac{Q}{\left(h^2+a^2\right)}cos\theta=mg$$

$$Q=\pi\in_0\frac{mg}{h}\left(h^2+a^2\right)^{\frac{3}{2}}$$

21. Mass remove $=\frac{M}{\frac{4}{3}\pi R^3}\times\frac{4}{3}\pi\left(\frac{R}{3}\right)^3=\frac{M}{27}$

$$0-\frac{2R}{3}.\frac{M}{27}=\left(M-\frac{M}{27}\right)x$$

$$0-\frac{2RM}{3\times27}=\frac{26}{27}x$$

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$$x = \frac{2R}{3\times 26} = \frac{R}{39}$$

22. $\vec{P}.\vec{Q} = 0 \Rightarrow m = 10$

23. $T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$= 300\left(2^5\right)^{\frac{7}{5}-1} = 1200$

24. From equation of trajectory

$$y = x\tan\theta - \left(\frac{g}{2u^2\cos^2\theta}\right)x^2$$

On putting $\theta = 60^0$ *and* $x = \frac{R}{3} = \frac{u^2}{3g}\sin 2\theta = \frac{u^2}{2\sqrt{3}g}$, we get

$$y = \sqrt{3}\times\frac{u^2}{2\sqrt{3}g} - \frac{g}{2u^2\times\frac{1}{4}}\times\left(\frac{u^2}{2\sqrt{3}g}\right)^2$$

$$= \frac{u^2}{2g} - \frac{u^2}{6g} = \frac{u^2}{3g}$$

Loss in KE = Gain in PE = mg y

$$= mg\times\frac{u^2}{3g} = \frac{mu^2}{3}$$

$$KE = \frac{1}{2}mu^2 - \frac{1}{3}mu^2 = \frac{1}{6}mu^2$$

25. $24 - (-18\times 2) - 10I = 0$

$$I = \frac{60}{10} = 6A$$

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CHEMISTRY

26. Conceptual

27. Here $n_2 = 5$ & $n_1 = 2$

So longest wavelength means least energy difference transition i.e $n_2 = 5$ to $n_1 = 4$

$$\frac{1}{\lambda} = R_H (1)^2 \left(\frac{1}{(4)^2} - \frac{1}{(5)^2} \right)$$

So $\lambda = 4 \times 10^{-6} m$

29. $\Delta T_f = K_f \times \text{molality} \times \text{i}$

$0.558 = 1.86 \times 0.1 \times i$

Or $i = 3$

Thus complex is $[CoCl.xNH_3].Cl_2$, since co-ordination number of Co is six, thus complex is $\left[Co(NH_3)_5 Cl \right] Cl_2$

30. $Cu^{2+} + 2e^- \rightarrow Cu \quad E^0 = 0.337V \quad \Delta G^0 = -2F \times 0.337$

$Cu^{2+} + e^- \rightarrow Cu^+ \quad E^0 = 0.153V \quad \Delta G^0 = -1F \times 0.153$

$Cu^{2+} + 2e^- \rightarrow Cu \quad E_1^0 = 0.337V \quad \Delta G_1^0 = -2F \times 0.337$

$Cu^+ \rightarrow Cu^{2+} + e^- \quad E_2^0 = 0.153V \quad \Delta G_2^0 = 1 \times F \times 0.153$

$Cu^+ + e^- \rightarrow Cu \quad E_3^0 = ? \quad \Delta G_3^0 = -1 \times F \times E_3^0$

$\Delta G_3^0 = \Delta G_1^0 + \Delta G_2^0$

$-1 \times E_3^0 = -2F \times 0.337 + F \times 0.153$

$E_3^0 = 2 \times 0.337 - 0.153 = 0.521V$

32. Conceptual

33. $W_{H_2SO_4} = 392mg = 392 \times 10^{-3} g$

$M_{H_2SO_4} = 98$

Left moles = total moles – removed moles $= \frac{392 \times 10^{-3}}{98} - \frac{1.204 \times 10^{21}}{6.022 \times 10^{23}}$

Left moles $= 4 \times 10^{-3} - 2 \times 10^{-3} = 2 \times 10^{-3}$ moles

34. $\Delta H_{vap} = 40850 J\ mol^{-1}, T_b = 373K$

$$\Delta S_{vap} = \frac{\Delta H_{vap}}{T_b} = \frac{40850 J\ mol^{-1}}{373K} = 109.5 K^{-1} mol^{-1}$$

$$\Delta S_{vap}\ per\ gram = \frac{109.5 J\ K^{-1}}{18 g\ mol^{-1}} = 6.083 J\ K^{-1} g^{-1}$$

Entropy change for 3.6 g water = 6.083 J $K^{-1} g^{-1} \times 3.6 g = 21.89 J\ K^{-1}$

35. Number of unpaired electrons = 3

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$\mu_{eff} = 3.9BM$

Type of hybridization $= sp^3d^2$

36. *Cis* compound optically active

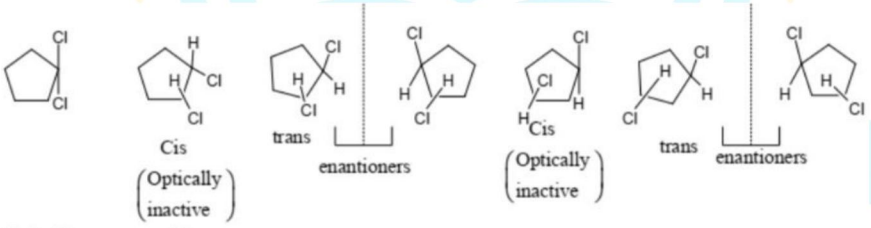
37.

$$PhCHO + CH_3CHO \rightarrow Ph-\underset{\underset{OH}{|}}{CH}-CH_2CHO$$

$$2CH_3CHO \rightarrow CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CHO$$

38. Electronegativity order: $O > N > S$

39.



Total isomers = 7

40. The product is

$$Cl-CH_2-\underset{\underset{Cl}{|}}{CH}-CH_2COCl$$

43.

$PbSO_4$, 0.02 M

(1) [NaI] = 0.1 M

$NaI \longrightarrow Na^+ + I^-$

0.1 M 0.1 M 0.1 M

(2) $Na_2SO_4 \longrightarrow 2Na^+ + SO_4^{-2}$

0.1 M 0.2 M 0.1 M

For precipitation of PbI_2,

$K_{sp}(PbI_2) = [Pb^{+2}][I^-]^2 = 9 \times 10^{-9} \Rightarrow [Pb^{+2}] \times (0.1)^2 = 9 \times 10^{-9}$

$\Rightarrow [Pb^{+2}]_{req} = 9 \times 10^{-7}$ M

For precipitation of $PbSO_4$,

$K_{sp}(PbSO_4) = [Pb^{+2}][SO_4^{-2}] = 1.8 \times 10^{-8} \Rightarrow [Pb^{+2}] \times 0.1 = 1.8 \times 10^{-8}$

$\Rightarrow [Pb^{+2}]_{req} = 1.8 \times 10^{-7}$ M

For precipitation of $PbSO_4$, required conc. of Pb^{+2} is less. **So, $PbSO_4$ precipitates first.**

When PbI_2 (more soluble compound) starts precipitating, $[Pb^{2+}] = 9 \times 10^{-7}$ M.

Then, conc. of anion of less soluble salt can be obtained as follows :

$K_{sp}(PbSO_4) = [Pb^{+2}][SO_4^{-2}] = 1.8 \times 10^{-8} \Rightarrow (9 \times 10^{-7})[SO_4^{-2}] = 1.8 \times 10^{-8}$

$[SO_4^{-2}]$ = **0.02 M**

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44. BCl_3 & $AlCl_3$ electron deficient molecule

XeF_2 super octet molecule

NO odd electron

IF_7 Super octet molecule

NO_2 odd electron

CIF_3 super octet molecule

CIO_2 odd electron

$CH_4, CO_2 \rightarrow octet$

45. 1^0 aliphatic & aromatic amines give carbyl amine test

46.

$K_b = 9 \times 10^{-7}$

pH = 10.48

pOH = 3.52

$[OH^-]$ = Antilog (–3.52) = Antilog $(1 - 0.48) \times 10^{-4} = 3 \times 10^{-4}$

BOH	$\rightleftharpoons$	B^+	+	OH^-
C		0		0
C – C α		C α		C α

$[OH^-] = C\alpha = 3 \times 10^{-4} \Rightarrow \alpha = 3 \times 10^{-3}$

$\alpha << 0.1$, $K_b = C\alpha^2 = 0.1 (3 \times 10^{-3})^2 = \mathbf{9 \times 10^{-7}}$

47. 233 g $BaSO_4$ contain 32g sulphur

0.4813g $BaSO_4$ contain $\frac{32 \times 0.4813}{233}$ g sulphur

% of sulphur $= \frac{32 \times 0.4813 \times 100}{233 \times 0.157} = 42.10$

48. $A =$ Benzoic acid

$B =$ Benzene

49.

75 atm

$CH_3OCH_3(g) \longrightarrow CH_4(g) + H_2(g) + CO(g)$

Pressure at t = 0 0.40

Pressure at t = 12 min (0.40 – P) P P P

For ideal gas behaviour moles ∝ pressure

$\therefore$ $a \propto 0.40$, $(a - X) \propto (0.40 - P)$

$\because$ $K = \frac{2.303}{t} \log \frac{a}{(a-X)}$

or $\frac{0.693}{14.5} = \frac{2.303}{12} \log \frac{0.40}{(0.40-P)}$

$\therefore$ P = 0.175 atm

Thus, total pressure after 12 minutes = 0.40 – P + P + P + P

= 0.40 + 2P

= 0.40 + 2 × 0.175 = **0.75 atm**

50. $\Delta G = \Delta H - T\Delta S$

$= 2.1 \times 10^3 - 20 \times 298 \quad = -3860 Cal$

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MATHEMATICS

51. $I=\int_{\pi/6}^{\pi/3}\frac{3\tan^2 x+6\tan x+11}{1+\tan^2 x}dx$

$I=\int_{\pi/6}^{\pi/3}\left(3\sin^2 x+6\sin x\cos x+11\cos^2 x\right)dx\ldots\ldots\ldots(i)$

Apply kings property

$I=\int_{\pi/6}^{\pi/3}\left(3\cos^2 x+6\cos x\sin x+11\sin^2 x\right)dx\ldots\ldots(ii)$

$(i)+(ii)$

$\Rightarrow 2I=\int_{\pi/6}^{\pi/3}(14+12\sin x\cos x)dx$

$\Rightarrow 2I=\int_{\pi/6}^{\pi/3}14+6(\sin 2x)dx$

$\Rightarrow 2I=14\left(\frac{\pi}{3}-\frac{\pi}{6}\right)+6\left[\frac{-\cos 2x}{2}\right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$

$\Rightarrow 2I=14\left(\frac{\pi}{6}\right)+3\left[\frac{1}{2}+\frac{1}{2}\right]$

$\Rightarrow I=\frac{7\pi}{3}+3$

$\Rightarrow I=\frac{7\pi+9}{6}=\frac{\pi a+b}{6}$

$\therefore a+b=16$

52. $\int\frac{dx}{(x-1)^{3/4}(x+2)^{5/4}}=\int\left(\frac{x+2}{x-1}\right)^{3/4}\frac{1}{(x+2)^2}dx$

Let $\frac{x-1}{x+2}=t$

$=\frac{1}{3}\int\frac{dt}{t^{3/4}}=\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{\frac{1}{4}}+c$

53. $k=\lim_{x\to 0^+}\frac{(2e)^x-x-x\ln 2-1}{x^2\frac{\tan x}{x}}$

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$$= \lim_{x\to 0^+} \frac{1 + x\ln(2e) + \frac{x^2}{2!}(\ln 2e)^2 + \ldots + x - x\ln 2 - 1}{x^2 \frac{\tan x}{x}}$$

$$= \frac{1}{2}(\ln 2 + 1)^2 = \frac{1}{2}(\ln 2)^2 + \ln 2 + \frac{1}{2}$$

54. The required equation of parabola is

$$\left(x - \frac{8}{9}\right)^2 = \frac{4}{9}\left(y - \frac{2}{9}\right)$$

Vertex of the parabola $\left(\frac{8}{9}, \frac{2}{9}\right)$

Length of latus recturm is $\frac{4}{9}$

$\therefore a = 8, b = 2, c = 4$

$a.b.c. = 64$

55. $\frac{dy}{dx} = \frac{1}{x\cos y + 2\sin y\cos y}$

$$\frac{dx}{dy} = x\cos y + 2\sin y\cos y$$

$$\frac{dx}{dy} + (-\cos y)x = 2\sin y\cos y$$

$$I.F = e^{\int -\cos y\, dy} = e^{-\sin y}$$

The solution is $x.e^{-\sin y} = 2\int e^{-\sin y}\sin y\cos y\, dy$

$$= -2\sin y.e^{-\sin y} - 2\int\left(-e^{-\sin y}\right)\cos y\, dy$$

$$= -2\sin y.e^{-\sin y} + 2\int e^{-\sin y}\cos y dy$$

$$= -2\sin y.e^{-\sin y} - 2.e^{-\sin y} + c$$

$$x = -2\sin y - 2 + c.e^{\sin y}$$

$$x = c.e^{\sin y} - 2(1 + \sin y)$$

56. $gof(x) = x$

$$g'f(x).f'(x) = 1$$

$$g'(f(x)) = \frac{1}{f'(x)}$$

$$g'(2) = \frac{1}{f'(0)}$$

$$g'(2) = 2$$

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$8g'(2)=16$

57. Using $P.V\ \bar{a}+2\bar{b}+3\bar{c}=0\rightarrow\bar{b}=\dfrac{-3\bar{a}}{2}-\dfrac{\bar{a}}{2}$

$\Delta ABC=\dfrac{1}{2}\left|3\left(\bar{c}\times\bar{a}\right)\right|$

And $\Delta AOC\dfrac{1}{2}\left|\bar{c}\times\bar{a}\right|\Rightarrow$ required ratio = 3

58. Conceptual

59. Given line $\dfrac{x+6}{-8}=\dfrac{y-7}{12}=\dfrac{z+5}{-16}=\lambda$

The coordinates of $Q=(2\lambda-6,7-3\lambda,4\lambda-5)$

Now $\overline{QR}=(2\lambda-7,-3\lambda,4\lambda-11)$

Since $\overline{QR}$ is perpendicular to the line

$\left(d.r's\ of\ \overline{QR}\right).\left(d.r's\ of\ line\right)=0$

$\Rightarrow 4\lambda-14+9\lambda+16\lambda-44=0$

$\Rightarrow 29\lambda=58$

$\Rightarrow\lambda=2$

$\therefore Q=(-2,1,3)$

$PQ=13$

60. Let $|z|=t,t\geq 0$

$e^{\frac{(t+3)(t-1)}{t+1}\log_e 2}\geq\log\sqrt{256}=8$

$2^{\frac{(t+3)(t-1)}{t+1}}\geq 2^3$

$\dfrac{(t+3)(t-1)}{t+1}\geq 3$

$\Rightarrow t^2+2t-3\geq 3t+3$

$\Rightarrow t^2-t-6\geq 0$

$\Rightarrow (t-3)(t+2)\geq 0$

$t\in(-\infty,-2)\cup[3,\infty)(\because t\geq 0)$

$t\in[3,\infty)$

Least value =3

61. $4x\leq 5y$

$x=1\Rightarrow y=1,2,3,4,5$

$x=2\Rightarrow y=2,3,4,5$

$x=3\Rightarrow y=3,4,5$

$x=4\Rightarrow y=4,5$

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$x=5 \Rightarrow y=5$

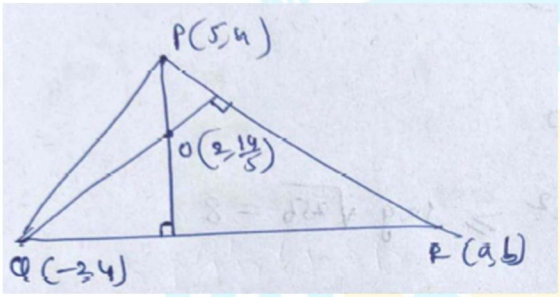
$$R=\left\{\begin{matrix}(1,1)(1,2),(1,3),(1,4),(1,5),(2,2),(2,3),(2,4),\\(2,5),(3,3),(3,4),(3,5),(4,4),(4,5),(5,5)\end{matrix}\right\}$$

To become R symmetric

Add $\{(2,1)(3,1)(4,1)(5,1)(3,2)(4,2)(5,2)(4,3)(5,3)(5,4)\}$

62. Conceptual

63.



Eqn of line $\overline{QR}$ is $5x+2y+2=0$

Eqn of line $\overline{PR}$ is $10x-3y-38=0$

$\therefore R=(2,-6)=(ab)$

$2a-b=10$

64. $\left(\sin^{-1}x-\cos^{-1}x\right)\left(\sin^{-1}x\right)^2+\left(\cos^{-1}x\right)^2+2\sin^{-1}x\left(\cos^{-1}x\right)=\dfrac{\pi^3}{16}$

$\Rightarrow \sin^{-1}x-\cos^{-1}x=\dfrac{\pi}{4}$

$\Rightarrow \cos^{-1}x=\dfrac{\pi}{8}$

$\Rightarrow x=\cos\dfrac{\pi}{8}$

65. $p^2=q^2+r^2+2\sqrt{q^4\cos^2\theta\sin^2\theta+r^4\cos^2\theta\sin^2\theta+q^2r^2\left(\cos^4\theta+\sin^4\theta\right)}$

$p^2=q^2+r^2+2\sqrt{q^4\cos^2\theta\sin^2\theta+r^4\cos^2\theta\sin^2\theta+q^2r^2\left(1-2\cos^2\theta\sin^2\theta\right)}$

$p^2=q^2+r^2+2\sqrt{\left(q^4+r^4-2q^2r^2\right)\cos^2\theta\sin^2\theta+q^2r^2}$

$p^2=q^2+r^2+2\sqrt{\left(q^2-r^2\right)^2\cos^2\theta\sin^2\theta+q^2r^2}$

$p^2=q^2+r^2+2\sqrt{\left(\dfrac{q^2-r^2}{2}\right)^2\left(\sin^2 2\theta\right)+q^2r^2}$

$\min\left(p^2\right)=q^2+r^2+2pr$

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$$\max\left(p^2\right)=q^2+r^2+2\sqrt{\left(\frac{q^2-r^2}{2}\right)^2+q^2r^2}$$

$$=q^2+r^2+\sqrt{\left(q^2-r^2\right)^2+4q^2r^2}$$

$$=q^2+r^2+q^2+r^2$$

$$=2\left(q^2+r^2\right)$$

$$\left|\max-\min\right|=\left|q^2+r^2-2qr\right|=\left(q-r\right)^2$$

66. $n_1=5\quad \overline{x_1}=5\quad \sigma_1^2=12$

$n_2=5\quad \overline{x_2}=8\quad \sigma_2^2=20$

New $\overline{x_1}=5-3=2$

New $\overline{x_2}=8+2=10$

Variance does not charge for new set

Mean of $c\left(\overline{x}\right)=\dfrac{n_1\overline{x_1}+n_2\overline{x_2}}{n_1+n_2}=\dfrac{10+250}{10}=6$

Combined variance $\left(\sigma\right)^2$

$$=\frac{n_1\sigma_1^2+n_2\sigma_2^2+n_1\left(\overline{x}-\overline{x_1}\right)^2+n_2\left(\overline{x}-\overline{x_2}\right)^2}{n_1+n_2}$$

$$=\frac{\left(5\times12\right)+\left(5\times20\right)+5\left(6-2\right)^2+5\left(10-6\right)^2}{10}$$

$$=32$$

$$\therefore \overline{x}+\sigma^2=6+32=38$$

67. $\dfrac{1}{6}\left|\left(\overline{AB}\times\overline{AC}\right).\overline{AD}\right|=\dfrac{1}{6}\left|-36\right|=6$

68. $\left|3adj\left(2A^{-1}.adj\left(2A^{-1}\right)\right)\right|$

$$=27\left|adj\left(2A^{-1}.adj\left(2A^{-1}\right)\right)\right|$$

$$=27\left|2A^{-1}.adj\left(2A^{-1}\right)\right|^2$$

$$=27\left(2^3\right)^2\left|A^{-1}.adj\left(2A^{-1}\right)\right|^2$$

$$=3^3.2^6\left|A^{-1}\right|^2.\left|adj\left(2A^{-1}\right)\right|^2$$

$$=3^3.2^6\frac{1}{\left|A\right|^2}.\left|2A^{-1}\right|^4$$

$$=3^3.2^6.\frac{1}{2^2}.2^{12}.\frac{1}{\left|A\right|^4}$$

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$= 3^3.2^4.2^{12}.\frac{1}{2^4}$

$= 3^3.2^{12}$

69. $\begin{vmatrix} \lambda & \lambda+1 & \lambda-1 \\ \lambda+1 & \lambda & \lambda+2 \\ \lambda-1 & \lambda+2 & \lambda \end{vmatrix} = 0$

$C_2 \to C_2 - C_1$

$C_3 \to C_3 - C_1$

$\Rightarrow \begin{vmatrix} \lambda & \lambda+1 & \lambda-1 \\ 1 & -1 & 3 \\ -1 & 1 & 1 \end{vmatrix} = 0$

$\Rightarrow \lambda(-1-3)-(\lambda+1)(1+3)+(\lambda-1)(1-1)=0$

$\Rightarrow -4\lambda-4\lambda-4=0 \qquad \Rightarrow 8\lambda+4=0 \Rightarrow \lambda=\frac{-1}{2}$

70. Conceptual

71. $f(x) = \max\{|x+1|,|x+2|,|x+3|,|x+4|,|x+5|\}$

Since, area will be maximum from -5 to 1

Thus, from, graph

$\int_{-6}^{0} f(x)\,dx = \int_{-6}^{-3}|x+1|\,dx + \int_{-3}^{0}|x+5|\,dx$

$= \int_{-6}^{-3} -(x+1)\,dx + \int_{-3}^{0}(x+5)\,dx \qquad = -\left[\frac{x^2}{2}+x\right]_{-6}^{-3} + \left[\frac{x^2}{2}+5x\right]_{-3}^{0}$

$= -\left[\left(\frac{9}{2}-3\right)-(18-6)\right]+\left[0-\left(\frac{9}{2}-15\right)\right] \qquad = -\left[\frac{3}{2}-12\right]+\frac{21}{2}=21$

72. $\left(x^3+5x^2-\frac{2}{x^5}\right)^{10} = \sum \frac{10!}{r_1!r_2!r_3!}.(x^3)^{r_1}.(5x^2)^{r_2}\left(\frac{-2}{x^5}\right)^{r_3}$

$= \sum \frac{10!}{r_1!r_2!r_3!}(-2)^{r_3}(5)^{r_2}(x)^{3r_1+2r_2-5r_3}$

To get constant term $3r_1+2r_2-5r_3=0$

$3r_1+2r_2=5r_3$

$\therefore$ constant term $= \frac{10!}{6!3!}(5)^6(-2)^3 \qquad = -2^6.3^1.5^7.7^1$

73. $x = 2+m$

$my = 4-m \Rightarrow (x-2)(y+1)=4$

$XY = 4$

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Given circle $(x-2)^2+(y+1)^2=25$

$X^2+Y^2=25$

Hence both are concentric curves

$\therefore 0P^2+0Q^2+0R^2+0S^2=4r^2$

$=4(25)=100$

74. $f^1(x)=\dfrac{(x^2-6x-6)-(4)(2x-6)}{(x^2-6x-16)^2}$

$=\dfrac{-x^2-16}{(x^2-6x-16)^2}$

$f^1(x)<0$

f is decreasing in

$(-\infty,-2)\cup(-2,8)\cup(8,\infty)$

Hence $a=2\ b=8\ a+b=10$

75. Let $y=\sqrt{4-\dfrac{1}{3\sqrt{2}}\sqrt{4-\dfrac{1}{3\sqrt{2}}\sqrt{4-\dfrac{1}{3\sqrt{2}}}.....}}$

$\therefore y=\sqrt{4-\dfrac{1}{3\sqrt{2}}y}$

$y^2=4-\dfrac{1}{3\sqrt{2}}y$

$y^2+\dfrac{1}{3\sqrt{2}}y-4=0$

$3\sqrt{2}y^2+y-12\sqrt{2}=0$

$y=\dfrac{-1\pm17}{6\sqrt{2}}$ or $y=\dfrac{8}{3\sqrt{2}}$

Now, $6+\log_{3/2}\left(\dfrac{1}{3\sqrt{2}}y\right)=6+\log_{3/2}\left(\dfrac{1}{3\sqrt{2}}\cdot\dfrac{8}{3\sqrt{2}}\right)$

$=6+\log_{3/2}\left(\dfrac{4}{9}\right)$

$=6+\log_{3/2}\left(\dfrac{3}{2}\right)^{-2}$

$=6-2=4$

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